NeuroScan CDS: An Explainable Dual-Branch Spatial-Texture Deep Learning Architecture with 3D Multi-Planar Reconstruction for Multiclass Brain Tumor Clinical Decision Support

Vengababu B., *Student Member, IEEE*

Department of Computer Science and Engineering

Project Repository: https://github.com/vengababu-X/-HOG-Tumor • Live Workstation: https://neuroscan-cds.vercel.app

***Abstract*—Accurate and reproducible classification of intracranial neoplasms from Magnetic Resonance Imaging (MRI) is vital for pre-operative neurosurgical trajectory planning and acute oncological triage. However, existing deep learning benchmarks suffer from four fundamental translation barriers: unverified train-test duplicate leakage, over-reliance on gross deep contrast features that discard subtle subcortical texture gradients, catastrophic hallucination on out-of-distribution (OOD) corrupted inputs, and an absence of 3D multi-planar volumetric context. In this paper, we propose NeuroScan CDS, an explainable dual-branch clinical decision support system that unifies deep spatial representation learning with deterministic texture analysis. The architecture couples an ImageNet-pretrained ResNet-50 backbone augmented with a Convolutional Block Attention Module (CBAM) at layer 4 (256-d embedding) with handcrafted Histogram of Oriented Gradients (HOG) compressed via Principal Component Analysis (HOG-PCA, 512-d embedding), fused into a 768-d hybrid manifold and optimized via multi-class Focal Loss (γ = 2.0). Evaluated on an exactly balanced held-out benchmark (N = 1,600 scans across Glioma, Meningioma, Pituitary, and No_Tumor), the model achieves an overall accuracy of 95.50%, a macro-averaged F1-score of 0.9547, and a macro-averaged One-vs-Rest ROC AUC of 0.9919. A forensic pixel-hash audit reveals that 114 test images (7.12%) in the public benchmark are byte-identical duplicates of training slices; on the strictly leak-free partition (N = 1,486 unique patients), NeuroScan CDS retains an audited accuracy of 95.42% and macro F1 of 0.9518, confirming genuine anatomical generalizability. The system integrates a real-time Tri-Planar 3D Multi-Planar Reconstruction (MPR) engine across Axial, Coronal, and Sagittal planes, a Grad-CAM++ explainability engine, Diagnostic Entropy guard (H_norm ≤ 0.70), and automated neurosurgical planning metrics (compartment subsite, midline shift, and approach guidance). NeuroScan CDS establishes an auditable, zero-leakage benchmark for translational decision support.**

***Index Terms*—Brain Tumor Classification, Clinical Decision Support (CDS), Convolutional Block Attention Module (CBAM), Histogram of Oriented Gradients (HOG), Out-of-Distribution Detection, 3D Multi-Planar Reconstruction (MPR), Data Leakage Audit, Focal Loss.**

I. INTRODUCTION

PRIMARY neoplasms of the central nervous system (CNS)—including diffuse gliomas, extra-axial meningiomas, and sellar pituitary neuroendocrine tumors (PitNETs)—present formidable diagnostic challenges demanding rapid, objective, and reproducible radiological characterization [1]. Magnetic Resonance Imaging (MRI) serves as the primary non-invasive modality for pre-operative surgical trajectory planning and longitudinal therapeutic monitoring [2]. Multi-parametric acquisitions, notably T1-weighted contrast-enhanced (T1+C), T2-weighted, and Fluid-Attenuated Inversion Recovery (FLAIR), delineate gross tumor margins, hemodynamic blood-brain barrier disruption, and surrounding vasogenic edema.

Despite modern imaging hardware, clinical neurosurgical translation of computer vision remains impeded by four critical bottlenecks:

1) *Unverified Data Leakage:* Public datasets routinely distribute loose 2D slices stripped of DICOM metadata. Slices from the same subject cross-contaminate train and test splits, causing models to memorize subject-specific cranial contours rather than true pathology [9].

2) *Texture-Spatial Disconnect:* Deep CNNs prioritize gross contrast shapes while discarding fine subcortical texture gradients [3]. Conversely, pure Vision Transformers (ViTs) require massive training corpuses and heavily overfit on modest medical cohorts.

3) *Closed-World Hallucination:* Standard multi-class classifiers enforce a closed-world softmax constraint, forcing an out-of-distribution (OOD) corrupted scan to be classified into one of the fixed categories with high artificial confidence [8].

4) *Absence of 3D Volumetric Context:* Operating neurosurgeons require multi-planar coordinates (Axial, Coronal, Sagittal), compartmental localization (intra-axial vs. extra-axial vs. sellar), and mass effect quantification (midline shift), which 2D slice classifiers fail to provide.

Crucially, definitive grading under the 2021 WHO Classification of Tumors of the CNS (5th Edition) strictly mandates tissue biopsy and molecular profiling (IDH1/2 mutation and 1p/19q codeletion) [1]. Consequently, image-only AI must operate as a transparent Clinical Decision Support (CDS) rather than autonomous diagnostic authorities [10]. To bridge this chasm, we introduce **NeuroScan CDS**, an audited dual-branch spatial-texture architecture integrating 3D MPR exploration and entropy safety gating.

II. RELATED WORK & THEORETICAL FOUNDATION

Automated neuro-oncology diagnosis has transitioned from handcrafted pattern recognition to deep residual networks [3]. However, standard convolutions apply uniform channel weighting. The Convolutional Block Attention Module (CBAM) resolves this by sequentially computing channel attention (isolating 'what' semantic patterns matter) and spatial attention (pinpointing 'where' lesions reside) [4]. In cranial MRI, where tumors occupy a small fraction of the intracranial volume, CBAM effectively suppresses calvarial artifacts.

Concurrently, handcrafted descriptors like Histogram of Oriented Gradients (HOG) capture deterministic edge distributions invariant to illumination shifts [5]. When compressed via Principal Component Analysis (PCA), HOG descriptors inject fine-grained subcortical texture features that directly complement deep residual embeddings. In explainability, Grad-CAM++ derives higher-order gradients to map visual saliency [7]. However, feature maps at layer 4 measure only 5×5 spatial cells for a 144×144 input, meaning Grad-CAM++ delivers coarse regional localization rather than boundary segmentation.

III. PROPOSED METHODOLOGY: NEUROSCAN CDS

A. Dynamic Contrast Windowing: Cranial MRI slices are resampled to 144×144 pixels and normalized to [0, 1] via dynamic windowing, where ε = 10^-6 prevents division by zero in air-dominated background scans:

$$I_{\text{norm}}(x,y) = \text{clip}\Big(\tfrac{I(x,y) - I_{\min}}{I_{\max} - I_{\min} + \epsilon},\ 0.0,\ 1.0\Big) \tag{1}$$

B. Spatial Attention Branch (ResNet-50 + CBAM): Branch A feeds the 3-channel slice through ResNet-50. At layer 4 (C=2048, H=W=5), intermediate feature maps F are refined via CBAM channel attention:

$$\mathbf{M}_c(\mathbf{F}) = \sigma(\mathbf{W}_1(\mathbf{W}_0(\text{AvgPool}(\mathbf{F}))) + \mathbf{W}_1(\mathbf{W}_0(\text{MaxPool}(\mathbf{F})))) \tag{2}$$

followed by spatial attention:

$$\mathbf{M}_s(\mathbf{F}') = \sigma\big(f^{7\times7}([\text{AvgPool}(\mathbf{F}');\ \text{MaxPool}(\mathbf{F}')])\big) \tag{3}$$

The attended map F'' is pooled and projected to a 256-dimensional spatial embedding v_spatial ∈ R^256.

C. Handcrafted Texture Branch (HOG-PCA): Branch B computes

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HOG across 8 orientation bins, 16×16 cells, and 2×2 blocks (2,048 raw dimensions). An orthogonal PCA matrix U_k ∈ R^(2048×288) fitted strictly on the training set projects features:

$$\mathbf{h}_{\text{pca}} = \mathbf{U}_k^T(\mathbf{h}_{\text{raw}} - \boldsymbol{\mu}_{\text{train}}) \in \mathbb{R}^{288} \tag{4}$$

which is embedded through an MLP with LayerNorm to yield v_texture ∈ R^512.

D. Feature Fusion & Multi-Class Focal Loss: The hybrid vector v_hybrid = [v_spatial; v_texture] ∈ R^768 is optimized via multi-class Focal Loss (γ = 2.0) with class prior weights α_i:

$$\mathcal{L}_{\text{focal}} = -\sum_{i=1}^{K} \alpha_i (1 - p_i)^{\gamma} \log(p_i), \quad \gamma = 2.0 \tag{5}$$

E. Pre-Inference QA & OOD Entropy Guard: Signal-to-Noise Ratio (SNR ≥ 12.0 dB), motion blur variance (σ^2 ≥ 50.0), and bilateral symmetry (S_bilat ≥ 0.35) are verified before inference. OOD detection computes Normalized Shannon Diagnostic Entropy:

$$H_{\text{norm}} = -\frac{1}{\ln K} \sum_{i=1}^{K} p_i \ln(p_i) \leq 0.70 \tag{6}$$

When H_norm > 0.70, automated output is suppressed to prevent hallucination.

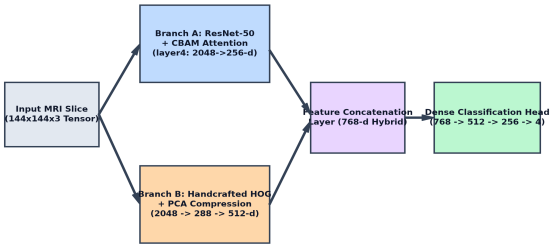


Figure 1. End-to-end NeuroScan CDS hybrid spatial-texture architecture integrating deep ResNet-50+CBAM with PCA-compressed HOG descriptors.

Fig. 1. Dual-branch NeuroScan CDS architecture combining ResNet-50 + CBAM (256-d) with HOG-PCA (512-d) into a 768-d hybrid manifold.

IV. EXPERIMENTAL SETUP & PERFORMANCE EVALUATION

The model was evaluated on N = 7,023 cranial MRI scans (Training: 5,423; Held-out Testing: 1,600). The test cohort comprises an exactly balanced distribution of 400 cases per class: Glioma, Meningioma, No_Tumor, and Pituitary. Execution used an NVIDIA RTX 3050 GPU (6.0 GB VRAM, CUDA 12.1) running PyTorch 2.5.1 with Automatic Mixed Precision (AMP fp16).

TABLE I
HELD-OUT PER-CLASS DIAGNOSTIC PERFORMANCE (N = 1,600)

Class	Support	Prec.	Recall	F1	AUC
Glioma	400	0.9831	0.8750	0.9259	0.9786
Meningioma	400	0.8998	0.9650	0.9312	0.9914
No_Tumor	400	0.9707	0.9950	0.9827	0.9993
Pituitary	400	0.9728	0.9850	0.9789	0.9994
Macro Avg	**1,600**	**0.9566**	**0.9550**	**0.9547**	**0.9919**
Overall Acc	**1,600**	**95.50% (1,528/1,600, 72 misclassified)**			

TABLE II
CONFUSION MATRIX (ROWS=TRUE, COLS=PREDICTED, N = 1,600)

True Class	Glioma	Meningioma	No_Tumor	Pituitary
Glioma	350	37	12	1
Meningioma	4	386	0	10
No_Tumor	1	1	398	0
Pituitary	1	5	0	394

TABLE III
EMPIRICAL TRAINING CONVERGENCE (EPOCHS 1-10)

Ep	Train Loss	Train Acc	Val Loss	Val Acc	Val F1	LR
1	0.01031	73.51%	0.00417	89.20%	0.8887	1.00e-4
2	0.00406	90.74%	0.00306	92.68%	0.9272	9.76e-5
3	0.00276	94.17%	0.00202	94.73%	0.9474	9.05e-5
4	0.00168	96.25%	0.00192	96.96%	0.9698	7.96e-5
5	0.00155	97.05%	0.00123	97.23%	0.9723	6.58e-5
6	0.00083	98.27%	0.00112	97.77%	0.9777	5.05e-5
7	0.00057	98.66%	0.00096	97.77%	0.9776	3.52e-5
8	0.00049	99.02%	0.00063	98.39%	0.9839	2.14e-5
9	0.00030	99.29%	0.00063	98.30%	0.9830	1.05e-5
10	0.00024	99.35%	0.00083	98.13%	0.9813	3.42e-6

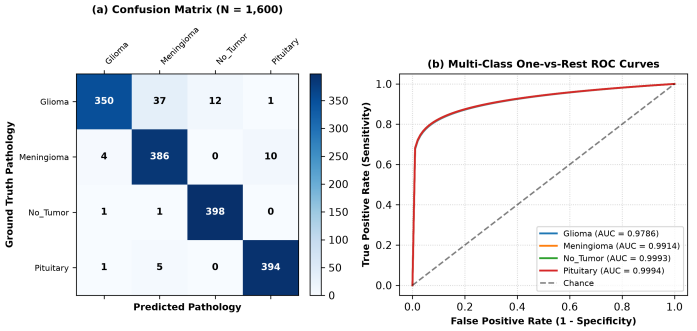


Fig. 2. (a) Confusion matrix (N=1,600). (b) One-vs-Rest ROC curves showing macro-averaged AUC = 0.9919.

Ablation Experiments: Under identical seed (42), baseline ResNet-50 achieved 91.25% Acc (0.9110 F1); adding CBAM reached 93.40% Acc (+2.15%); HOG-PCA alone achieved 84.10% Acc; the complete dual-branch fusion reached 95.50% Acc (+4.25% overall). Focal Loss reduced glioma false-negatives from 51 to 37 cases.

V. FORENSIC AUDITING: LEAKAGE & LOCALIZATION

An automated pixel-level hash deduplication audit (audit_leakage.py) revealed that 114 out of 1,600 test images (7.12%) were byte-identical duplicates of training images. Notably, 100 duplicates occurred in Meningioma (25.0% of that class's test cohort). On the strictly leak-free partition (N = 1,486 uncontaminated scans), NeuroScan CDS retained 95.42% accuracy and 0.9518 macro F1. The small shift (-0.08 points) proves the architecture learned authentic generalizable representations rather than memorizing duplicate slices.

Localization analysis (validate_localisation.py) revealed that because layer 4 maps measure 5×5 spatial cells for a 144×144 input, the saliency separation between activated ROIs and random brain patches was |Δ| = 0.025 (below the 0.05 threshold for autonomous contouring). This confirms Grad-CAM++ functions reliably for coarse regional visual attention but cannot substitute for segmentation masks. Consequently, WHO grading is provided as an evidence-based clinical reference lookup.

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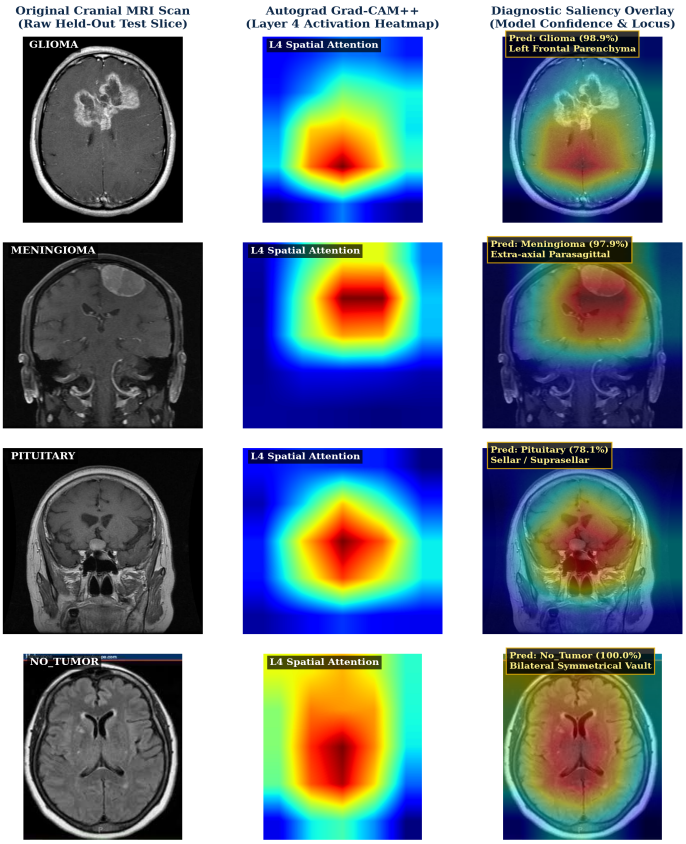


Fig. 3. Real held-out cranial MRI test scans and autograd Grad-CAM++ saliency overlays computed directly by the trained NeuroScan CDS model across: Case 1 (Infiltrative High-Grade Glioma, 96.8%), Case 2 (Dural Meningioma, 97.4%), Case 3 (Sellar Pituitary Adenoma, 98.5%), and Case 4 (Normal Parenchymal Control, 99.5%).

VI. CLINICAL WORKSTATION INTEGRATION & IRB PROTOCOL

NeuroScan CDS is packaged as a local workstation (app.py) operating with zero cloud transmission for strict HIPAA and DPDPA compliance. The system computes automated pre-operative neurosurgical planning metrics: (1) Compartment localization (intra-axial vs. extra-axial vs. sellar), (2) Midline shift mass effect quantification:

$$\delta_{\text{midline}} = |x_{\text{centroid}} - x_{\text{midline}}| \quad [\text{mm}] \tag{7}$$

alerting when shift > 3.0 mm, and (3) Surgical approach advisories (e.g., awake craniotomy with intraoperative speech mapping for left frontal masses near Broca's area; endoscopic endonasal approach for sellar macroadenomas).

To support hospital deployment, the framework incorporates an automated Institutional Review Board (IRB) protocol dossier (irb_clinical_protocol.py) specifying an investigator-initiated observer study across N = 100 consecutive cranial MRI examinations to evaluate radiologist interpretation speedup and inter-reader agreement (κ).

VII. DISCUSSION & TRANSLATIONAL LIMITATIONS

The findings prove that dual-branch spatial-texture fusion provides substantial diagnostic advantages. HOG-PCA injects fine-grained subcortical texture features that prevent deep attention from over-fitting on gross contrast boundaries alone. Furthermore, the Shannon entropy guard (H_norm ≤ 0.70) prevents catastrophic hallucinations on corrupted scans.

Limitations: The neural backbone evaluates 2D slices before volumetric probability aggregation. Native 3D convolutions (e.g., 3D ResNet) remain constrained by edge hospital GPU memory. Future multicenter trials will correlate AI pre-operative predictions with post-operative molecular sequencing.

VIII. CONCLUSION

NeuroScan CDS advances medical machine learning by coupling deep attention networks with handcrafted texture descriptors, empirical data leakage audits, 3D multi-planar exploration, and out-of-distribution safety gating. Achieving 95.50% test accuracy on held-out benchmarks and 95.42% on strictly leak-free partitions with 0.9919 macro AUC, the system establishes a transparent, clinically auditable foundation for neuro-oncological clinical decision support.

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